

Crossing-Point Analysis for Weighted Rolling Validation: Simulation Study

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1 Purpose and new model

This report checks the new Analysis 09 crossing formulas at finite training sizes. The theory report supplies the proofs. The model is $Y_i \stackrel{\text{iid}}{\sim} N(\theta, 1)$, with $0 < |\theta| < 1$: Model 0 uses the sample mean of the available training data, while Model 1 always predicts zero. Model 1 is not an oracle here. Negative loss differences favor Model 0.

The population comparison after n training observations is $\Delta_R(n) = 1/n - \theta^2$. Expected weighted rolling validation (RV) has the same sign as

$$\frac{\Delta_w(n)}{W_n} = \frac{\sum_{i=2}^{n+1} w_i / (i-1)}{\sum_{i=2}^{n+1} w_i} - \theta^2.$$

The division by $W_n > 0$ puts the curves on a comparable scale without changing a crossing. Ordinary RV is the unweighted reference; log, squared-log, and log-log weights come from 08; geometric weighting is a new 09 diagnostic for crossing alignment. The order is fixed throughout.

2 Design and checks

Exact formulas are evaluated for $|\theta| \in \{0.2, 0.1, 0.05, 0.025\}$, with training sizes through 40,000. The geometric base is 1.05. At each population crossing, we examine n/n_{pop} near 0.75, 1, 1.25, 2. The source checks the direct squared-loss score against a Gaussian quadratic form, and checks its exact mean and variance; the largest discrepancy is 1.7e-16.

A separate Monte Carlo check uses 2,000 Gaussian paths at $\theta = 0.1$, seed 2026, in batches of 200. Each path is reused across weights and training sizes. The code caches its output, so re-rendering does not repeat the experiment. Exact curves and Monte Carlo points are distinguished in every figure.

Table 1: Analysis 09 calculation and simulation design.

Item	Value
Gaussian mean magnitudes	0.2, 0.1, 0.05, 0.025
Exact maximum training size	40,000
Geometric base	1.05
Monte Carlo theta	0.1
Monte Carlo replications	2,000
Seed	2026
Batch size	200

3 Do expected RV and population risk cross together?

The theory predicts that ordinary RV crosses late. Logarithmic weighting should shorten this lag without eliminating it asymptotically; geometric weighting should move the expected RV crossing close to the population crossing. Figure 1 compares the exact population and normalized expected RV differences at $\theta = 0.1$. The grey dashed line marks the population crossing; the orange dotted line marks the expected RV crossing.

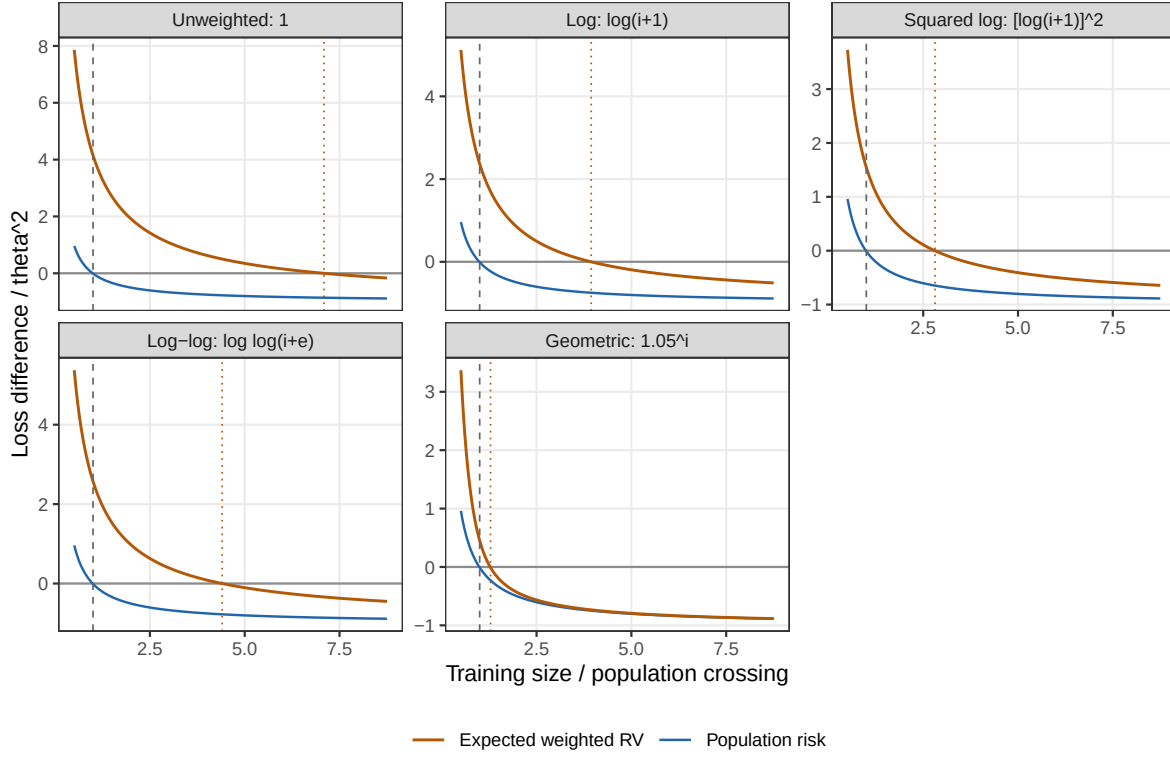


Figure 1: Exact population and expected weighted-RV loss differences at $\theta = 0.1$. Vertical lines mark their crossings.

Every weighted crossing is later than the population crossing. The log and squared-log weights cross earlier than ordinary RV, while geometric weighting is nearest at this parameter value. This is a crossing comparison, not a claim about random selection accuracy.

Figure 2 tests the asymptotic direction by shrinking $|\theta|$. The crossing ratio is one for exact alignment; a ratio above one measures lag.

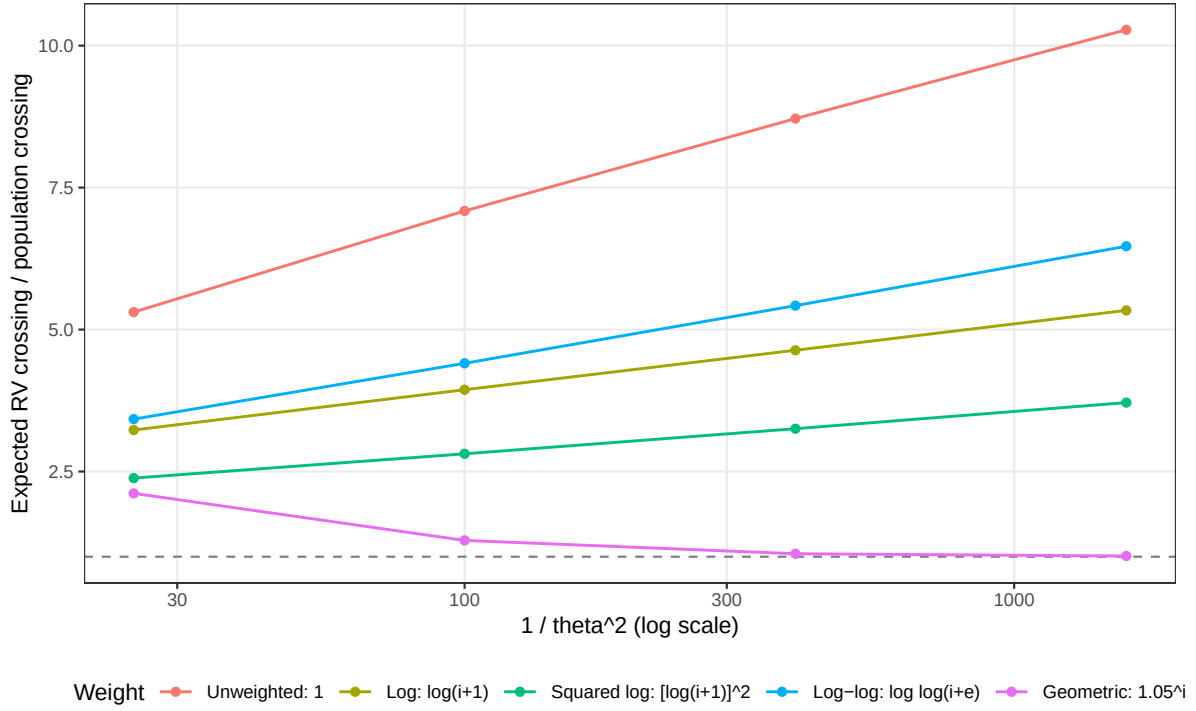


Figure 2: Exact expected-RV crossing ratios as the population crossing moves outward.

The geometric ratio approaches one over this grid. The logarithmic ratios rise, consistent with their theoretical logarithmic divergence. Faster growth within the two log-power examples reduces the lag's constant, not its asymptotic order.

Table 2: Exact crossing indices and improvement relative to ordinary RV.

theta	Weight	Population crossing	Expected RV crossing	Absolute difference	Crossing ratio	Improvement
0.200	Unweighted: 1	26	138	112	5.308	Reference
0.200	Log: $\log(i+1)$	26	84	58	3.231	Yes
0.200	Squared log: $[\log(i+1)]^2$	26	62	36	2.385	Yes
0.200	Log-log: $\log \log(i+e)$	26	89	63	3.423	Yes
0.200	Geometric: 1.05^i	26	55	29	2.115	Yes
0.100	Unweighted: 1	101	716	615	7.089	Reference
0.100	Log: $\log(i+1)$	101	398	297	3.941	Yes
0.100	Squared log: $[\log(i+1)]^2$	101	284	183	2.812	Yes
0.100	Log-log: $\log \log(i+e)$	101	445	344	4.406	Yes
0.100	Geometric: 1.05^i	101	130	29	1.287	Yes
0.050	Unweighted: 1	401	3495	3094	8.716	Reference
0.050	Log: $\log(i+1)$	401	1859	1458	4.636	Yes
0.050	Squared log: $[\log(i+1)]^2$	401	1305	904	3.254	Yes
0.050	Log-log: $\log \log(i+e)$	401	2174	1773	5.421	Yes
0.050	Geometric: 1.05^i	401	422	21	1.052	Yes
0.025	Unweighted: 1	1601	16458	14857	10.280	Reference
0.025	Log: $\log(i+1)$	1601	8546	6945	5.338	Yes
0.025	Squared log: $[\log(i+1)]^2$	1601	5946	4345	3.714	Yes
0.025	Log-log: $\log \log(i+e)$	1601	10354	8753	6.467	Yes
0.025	Geometric: 1.05^i	1601	1621	20	1.012	Yes

4 Does alignment retain enough averaging?

Analysis 09 proves that, near the moving crossing, the sufficient alignment condition $nA_n/W_n \rightarrow 1$ forces $n \sum w_i^2/W_n^2 \rightarrow \infty$. The new linear Gaussian variance term then prevents a vanishing Chebyshev ratio on either fixed side $n/n_{\text{pop}} \rightarrow c \neq 1$. Figure 3 uses the **exact** 09 variance, including both quadratic and linear parts, at three values of c . The ratio at $c = 1$ is omitted from this plot because the leading population difference is zero there.

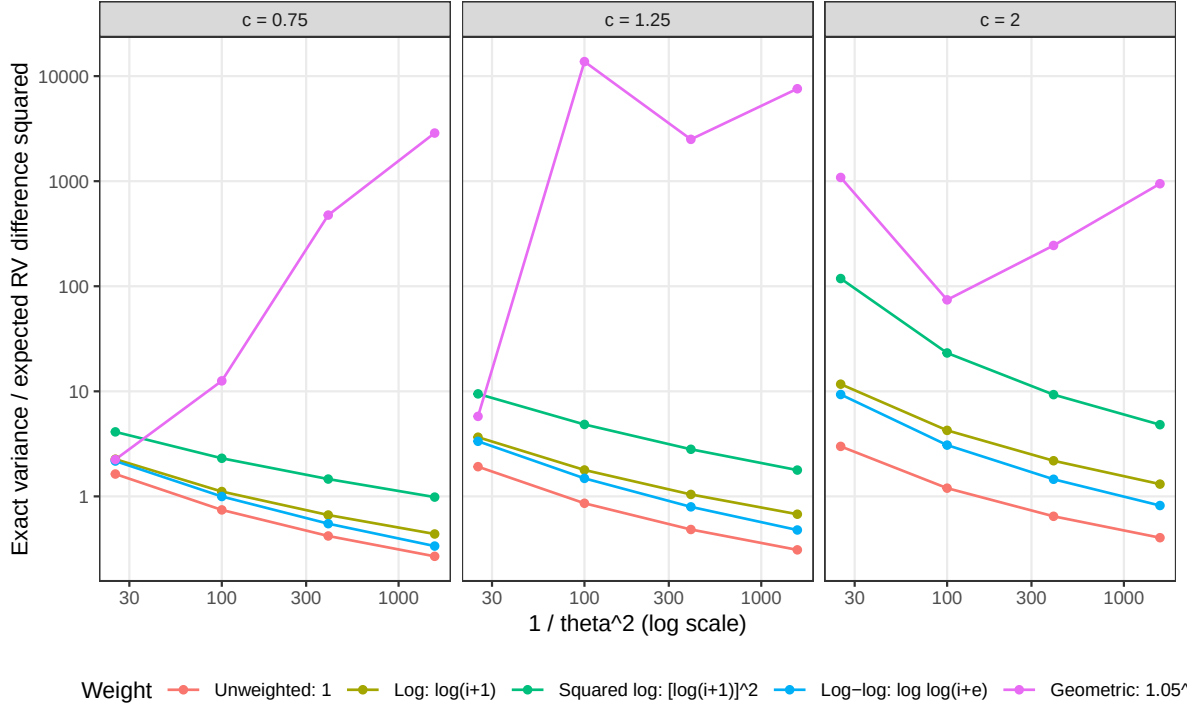


Figure 3: Exact Chebyshev ratios on both sides of the moving population crossing.

The log weights have small ratios over the shrinking- θ grid, but for $c > 1$ their expected RV comparison still favors the wrong model. Geometric weighting approaches crossing alignment, but its ratio stays large and can be irregular at finite sizes near its own crossing. This figure illustrates the two distinct failures identified by the theory; it does not establish a limiting selection probability.

The boundary $c = 1$ is shown separately in the table below. Its finite-grid ratios are descriptive: an asymptotic ratio with an exactly zero mean would be undefined.

Table 3: Exact near-crossing means and variance components at $\theta = 0.1$.

Target c	Weight	n	Population difference	Expected RV / W	Quadratic variance / W^2	Linear variance / W^2
0.75	Unweighted: 1	76	0.00316	0.05466	0.00170	0.00053
0.75	Log: $\log(i+1)$	76	0.00316	0.03324	0.00068	0.00055
0.75	Squared log: $[\log(i+1)]^2$	76	0.00316	0.02294	0.00061	0.00060
0.75	Log-log: $\log \log(i+e)$	76	0.00316	0.03549	0.00071	0.00054
0.75	Geometric: 1.05^i	76	0.00316	0.01312	0.00114	0.00102
1.00	Unweighted: 1	101	-0.00010	0.04146	0.00096	0.00040
1.00	Log: $\log(i+1)$	101	-0.00010	0.02369	0.00037	0.00041
1.00	Squared log: $[\log(i+1)]^2$	101	-0.00010	0.01546	0.00034	0.00045
1.00	Log-log: $\log \log(i+e)$	101	-0.00010	0.02573	0.00039	0.00041
1.00	Geometric: 1.05^i	101	-0.00010	0.00452	0.00086	0.00099
1.25	Unweighted: 1	126	-0.00206	0.03300	0.00062	0.00032
1.25	Log: $\log(i+1)$	126	-0.00206	0.01773	0.00023	0.00033
1.25	Squared log: $[\log(i+1)]^2$	126	-0.00206	0.01084	0.00021	0.00036
1.25	Log-log: $\log \log(i+e)$	126	-0.00206	0.01959	0.00024	0.00033
1.25	Geometric: 1.05^i	126	-0.00206	0.00035	0.00071	0.00098
2.00	Unweighted: 1	202	-0.00505	0.01915	0.00024	0.00020
2.00	Log: $\log(i+1)$	202	-0.00505	0.00826	0.00008	0.00021
2.00	Squared log: $[\log(i+1)]^2$	202	-0.00505	0.00359	0.00008	0.00022
2.00	Log-log: $\log \log(i+e)$	202	-0.00505	0.00975	0.00009	0.00020
2.00	Geometric: 1.05^i	202	-0.00505	-0.00439	0.00046	0.00098

5 Monte Carlo check: expectation versus realized selection

The same theoretical mean can be checked without treating simulated selection as a proof. Figure 4 compares exact normalized expected RV curves with Monte Carlo means from the shared paths.

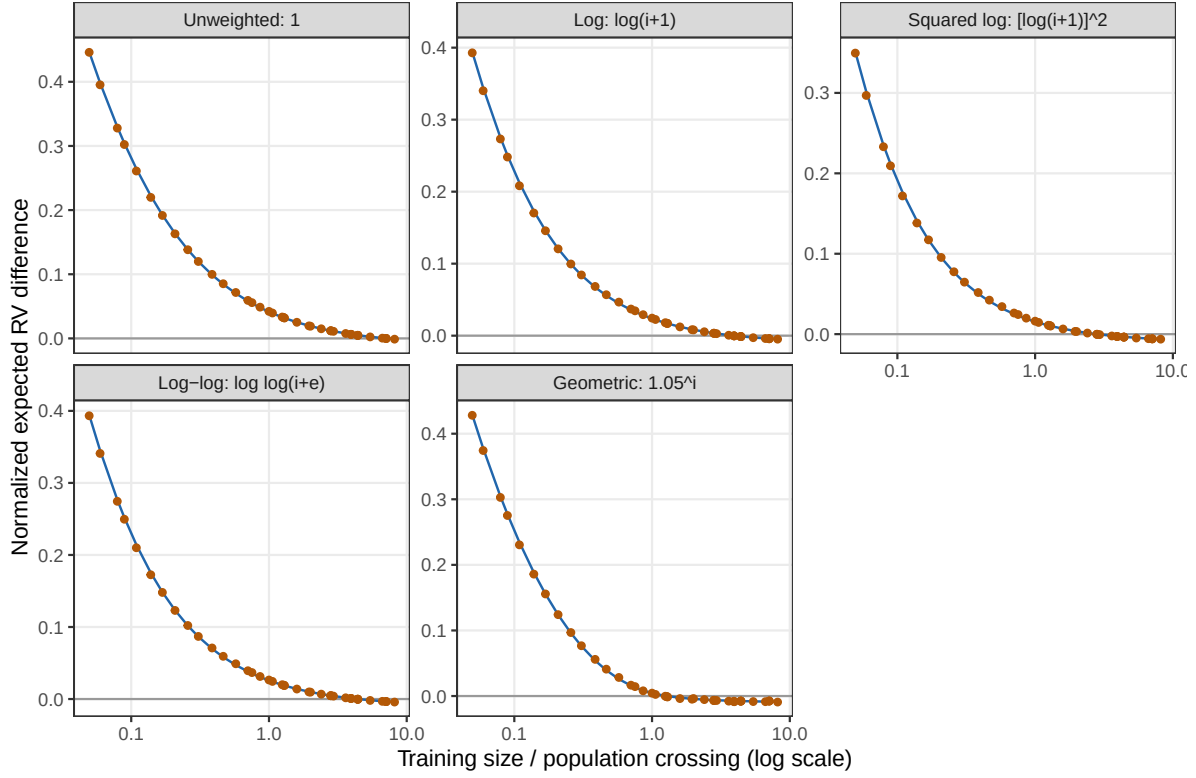


Figure 4: Exact normalized expected RV differences (blue) and Monte Carlo means (orange) at $\theta = 0.1$.

The empirical means follow the exact curves. Figure 5 asks a different question: how often does the realized RV score select Model 0? Pointwise bars use Monte Carlo standard errors; shared paths make comparisons across weights and sizes dependent.

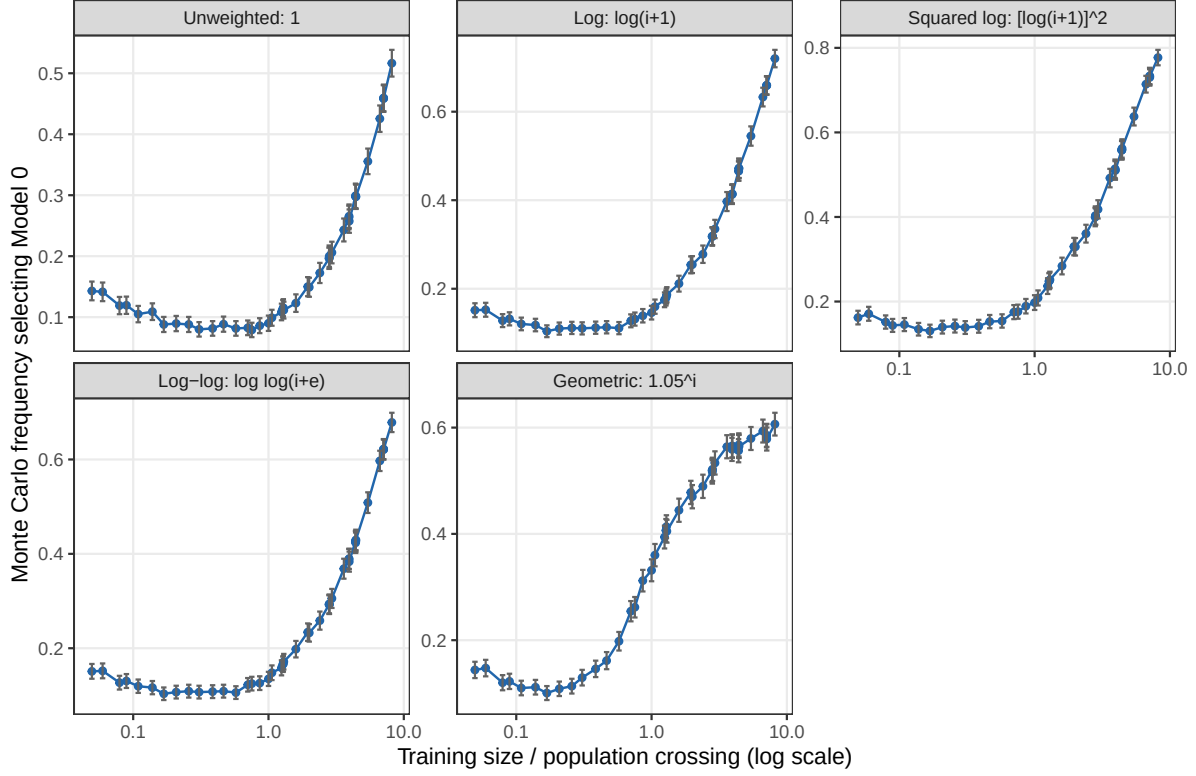


Figure 5: Monte Carlo frequency of selecting Model 0, with pointwise approximate 95% intervals.

For $n/n_{\text{pop}} > 1$, population risk favors Model 0. Expected sign alignment and a realized selection frequency near one are different requirements. The plotted frequencies describe this finite experiment; no asymptotic probability claim is inferred from them.

6 What these calculations establish

The exact crossing table and curves verify the finite-index formulas and illustrate the lag that general weighting changes. The exact variance calculations display the new linear noise term. Monte Carlo means support the expected RV calculation, while selection frequencies show the additional randomness faced by a realized criterion. The proofs and near-crossing impossibility statement are in the separate 09 theory report.